

Computer Programming 143 – Lecture 8

Program Control III

Faculty of Engineering



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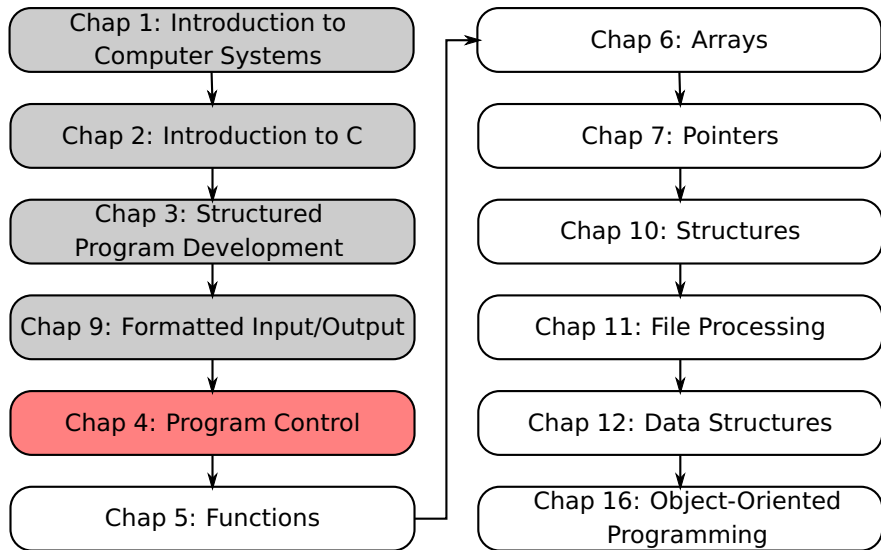
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Module Overview



Lecture Overview

- 1 The switch Multiple-Selection Statement (4.6)
- 2 The break Statement (4.8)
- 3 Summary of Structured Programming (4.11)

4.6 The **switch** Multiple-Selection Statement I

Unelegant **if** statement

```
if ( condition ) {  
    statement(s);  
}  
else if ( condition ) {  
    statement(s);  
}  
else if ( condition ) {  
    statement(s);  
}  
else if ( condition ) {  
    statement(s);  
}  
...
```

4.6 The **switch** Multiple-Selection Statement II

switch

- Useful when a variable or expression is tested for all the values it can assume and different actions are taken

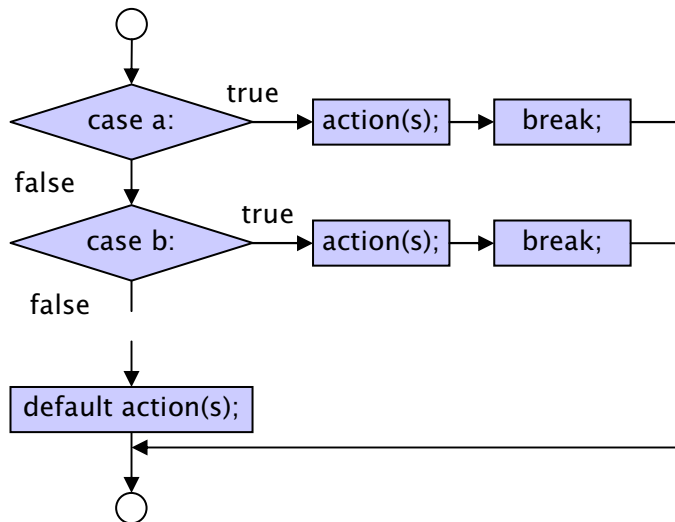
Format:

- Series of **case** labels and an optional **default** case

```
switch ( value ) {  
    case 1 : action(s);  
        break;  
    case 2 : action(s);  
        break;  
    default: action(s);  
}
```

- **break**; exits from statement

4.6 The **switch** Multiple-Selection Statement III



4.6 switch Simple Example I

```
int n_wheels;
printf("How many wheels does your vehicle have? ");
scanf("%d", &n_wheels);
switch (n_wheels)
{
    case 1:
        printf("It's a unicycle\n");
        break;
    case 2:
        printf("It might be a bicycle\n");
        break;
    case 3:
        printf("Probably a tricycle\n");
        break;
    case 4:
        printf("It might be a car\n");
        break;
    default:
        printf("No idea; we are far from solving AI\n");
        break;
}
```

4.6 `switch` Simple Example II

Example output

```
How many wheels does your vehicle have? 2  
It might be a bicycle
```

4.6 switch Bigger Example I

Example C code

```
/* Counting Grades (based on Fig. 4.5 in Deitel & Deitel) */
#include <stdio.h>
#include <stdlib.h>
// function main begins program execution
int main( void )
{
    int numStudents; //total number of students
    char grade; // grade input variable
    int passCount = 0; // number of passes
    int failCount = 0; // number of failures

    printf("Enter the total number of students.\n");
    scanf("%d",&numStudents); //read the total number of students
    printf( "Enter results (P for pass; F for failure).\n");
    do {
        scanf(" %c", &grade);//scanf reads one charater from keyboard
```

4.6 switch Bigger Example II

Example C code (cont'd...)

```
switch ( grade ) { // switch nested in while
  case 'F':        // grade input was uppercase F
  case 'f':        // or lowercase f
    failCount++; // increment number of failures
    break;       // exit switch

  case 'P':        // grade input was uppercase P
  case 'p':        // or lowercase p
    passCount++; // increment number of passes
    break;       // exit switch

  case '\n':       // ignore newlines
  case '\t':       // tabs,
  case ' ':        // and spaces in input
    break;       // exit switch
```

4.6 switch Bigger Example III

Example C code (cont'd...)

```
default:           // catch all other characters
    printf( "Invalid character entered. " );
    printf( "Enter a new result (P or F).\n" );
    break;
} // end switch
} while ((passCount+failCount) < numStudents); // end do....while

// output summary of results
printf( "\nTest results:\n" );
printf( "Passed: %d\n", passCount ); // display number of passes
printf( "Failed: %d\n", failCount ); // display number of failures

return 0; // indicate program ended successfully
} // end function main
```

4.6 **switch** Bigger Example IV

Example output

Enter the total number of students.

5

Enter results (P for pass; F for failure).

p

P

f

F

k

Invalid character entered. Enter a new result (P or F).

P

Test results:

Passed: 3

Failed: 2

4.8 The **break** Statement I

break

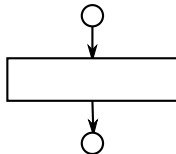
- Causes immediate exit from a **while**, **for**, **do...while** or **switch** statement
- Program execution continues with the first statement after the structure
- Common uses of the **break** statement
 - Escape early from a loop
 - Skip the remainder of a **switch** statement

Remarks

- Violates rules of structured programming when used with **while**, **for** or **do...while** statement
- For more information refer to the textbook

4.11 Summary of Structured Programming I

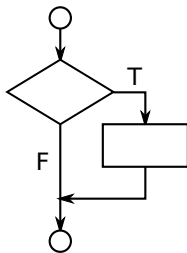
Sequence structure



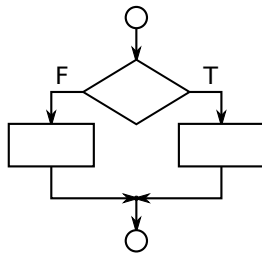
4.11 Summary of Structured Programming II

Selection structures

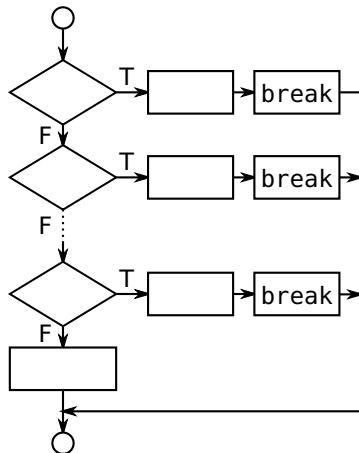
if statement



if...else statement



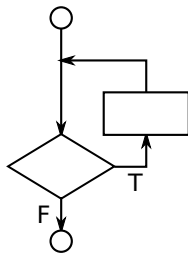
switch statement



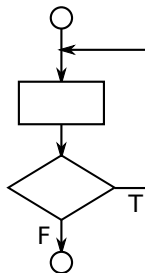
4.11 Summary of Structured Programming III

Repetition structures

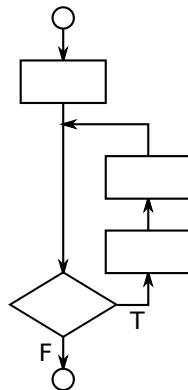
while statement



do...while statement

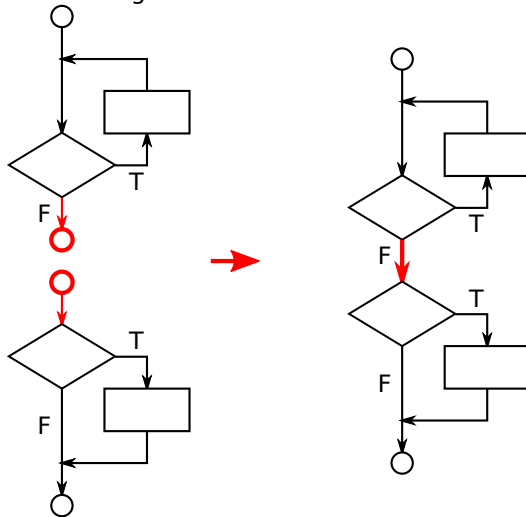


for statement



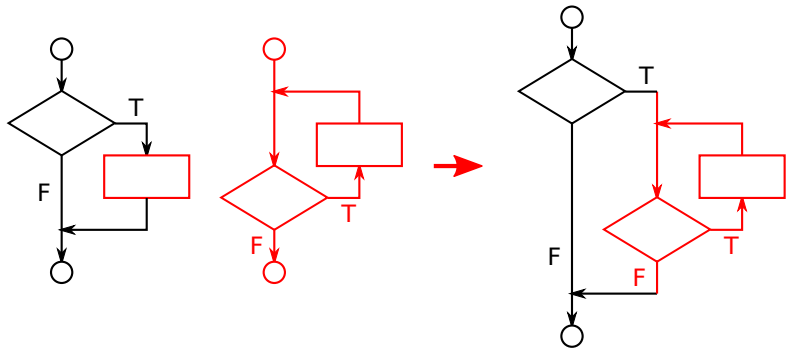
4.11 Summary of Structured Programming IV

Connecting control structures: method 1



4.11 Summary of Structured Programming V

Connecting control structures: method 2



Today

Program control III

- switch selection structure
- break and continue
- Summary of structured program development

Next lecture

Functions I

- Introduction to functions

Homework

- 1 Study Sections 4.6, 4.8, 4.11 in Deitel & Deitel
- 2 Do Self Review Exercises 4.1, 4.2(a)&(b), 4.4(c) in Deitel & Deitel
- 3 Do Exercises 4.5(b)&(c), 4.19, 4.24